

Travis Courtney

Associate Professor, University of Puerto Rico Mayagüez
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Appointments and professional experience:

2024–present	Associate Professor, University of Puerto Rico Mayagüez
2021–2024	Assistant Professor, University of Puerto Rico Mayagüez
2019–2021	Postdoctoral Scholar, Scripps Institution of Oceanography
2015–2019	National Science Foundation Graduate Research Fellow, Scripps Institution of Oceanography
2013–2014	Visiting Scientist, University of North Carolina at Chapel Hill
2013–2014	Laboratory Technician, Northeastern University
2013	Education Intern, North Carolina Aquarium at Pine Knoll Shores
2010–2013	Undergraduate Research Assistant, University of North Carolina at Chapel Hill

Education:

2019	Ph.D. Oceanography, Scripps Institution of Oceanography Dissertation: “Quantifying the rates and drivers of coral and coral reef calcification in the Anthropocene” Research Advisor: Dr. Andreas J. Andersson
2013	B.S. Geological Sciences + B.S. Environmental Sciences, University of North Carolina at Chapel Hill Thesis: “Impact of atmospheric $p\text{CO}_2$ and seawater temperature on the stable isotopic composition ($\delta^{18}\text{O}$, $\delta^{13}\text{O}$) of echinoid calcite (<i>Echinometra viridis</i>)” Research advisor: Dr. Justin B. Ries

Honors, scholarships, and awards:

2024	Sloan Research Fellowship in Earth System Science, Alfred P. Sloan Foundation
2020	Chancellor’s Dissertation Medal, University of California San Diego
2019	Graduate Student Excellence Research Award, Scripps Institution of Oceanography
2019	Travel Grant, University of California San Diego Graduate Student Association
2018	Graduate Student Excellence Travel Award, Scripps Institution of Oceanography
2018	Outstanding Mentor Award, Scripps Graduate Peer Mentor Program
2016	Outstanding Presentation Award, Scripps Student Symposium
2016	Scripps Fellowship, Scripps Institution of Oceanography
2015	Cody Fellowship, Scripps Institution of Oceanography
2015	Graduate Research Fellowship, National Science Foundation
2013	Graduated with Highest Honors and Distinction, UNC at Chapel Hill
2013	Environmental Excellence Award, UNC at Chapel Hill Institute for the Environment
2013	Carolina Research Scholar, UNC at Chapel Hill Office of Undergraduate Research
2013	1 st Place Undergraduate Poster Award, Anadarko Research Symposium
2012	Roy L. Ingram Geology Fund Scholarship, UNC at Chapel Hill Department of Geological Science
2012	Judson Mead Geologic Field Station Scholarship, Indiana University at Bloomington
2011	Harrington Scholar, UNC at Chapel Hill
2009	Wrightsville Beach Longboard Association Scholarship

Successful research grants and fellowships:

2024	Puerto Rico Sea Grant	\$210,001
	“Optimizing the cultivation of Ulva seaweed in the Caribbean: developing the framework for larger-scale cultivation and inclusion into the Puerto Rico Blue Economy”	
	PI: Dr. Stacey Williams, Dr. Travis Courtney, Dr. Alexandra Gregory, Manuel Olmeda Saldaña	
2024	Office of Naval Research	\$524,637
	“Seafloor classification and elevation changes on coral reefs in La Parguera, Puerto Rico using an autonomous surface vehicle”	
	PI: Dr. Travis Courtney	

2024	Office of Naval Research <i>This DEI Project was Terminated in Year 1 to Comply with Presidential Executive Order</i> "Inclusion and Belonging in Ocean Science (IBOS) Collaborative: The East Coast Division of the D-ENTERPRISE Initiative" PI: Dr. Travis Courtney , Dr. Andre Amador, Dr. Johana Rotterová	\$1,469,712 \$148,829
2024	Sloan Research Fellowship in Earth System Science PI: Dr. Travis Courtney	\$75,000
2024	United States Geological Survey "Influence of coral reef restoration on reef-growth capacity, wave transformation, and shoreline protection in Puerto Rico" PI: Dr. Andre Amador, Dr. Travis Courtney	\$319,999
2023	Office of Naval Research "Classification and structural complexity of marine substrates across spatial scales in Hawai'i using next-generation autonomous mapping vehicles" PI: Dr. Travis Courtney	\$208,025
2023	Schmidt Ocean Institute "To support four students' participation in R/V Falkor (too)'s research expedition FKt230417" PI: Dr. Travis Courtney	\$4,536
2022	Office of Naval Research Subaward from Purdue University "Blue Integrated Partnerships (BIP): Mentoring the Medici Scholars to Solve Tomorrow's Big Challenges" PI: Dr. Juan José Cruz Motta, Dr. Travis Courtney	\$61,946
2022	Puerto Rico Department of Natural and Environmental Resources "Development and Implementation of a Water Quality Monitoring Project in Shallow Coral Reef Areas Around Puerto Rico, and the Implementation of a 'BCG' Model" PI: Dr. Juan José Cruz Motta, Dr. Ernesto Weil, Dr. Travis Courtney	\$1,324,698
2022	EcoEléctrica "Quantifying net ecosystem metabolism and biogeochemical variability of seagrasses in Southwest Puerto Rico" PI: Dr. Travis Courtney , Catherine Hernandez Rodriguez [MS student in lab of PI Courtney]	\$94,925
2019	National Oceanographic and Atmospheric Administration "Quantifying coral reef net calcification capacity and vulnerability in the context of ocean acidification" PI: Dr. Andreas Andersson [Travis Courtney contributed significantly to proposal]	\$249,055
2018	National Science Foundation Biological Oceanography "Drivers of coral and reef-scale calcification in the North Atlantic" PI: Dr. Andreas Andersson [Travis Courtney contributed significantly to proposal]	\$265,466
2016	Shepard Foundation Student Fieldwork "Seasonal variability of net coral reef calcification in Kāne'ohe Bay, Hawai'i" PI: Travis Courtney	\$1,840
2016	Shepard Foundation Student Fieldwork "Impacts of widespread bleaching on net coral reef calcification in Kāne'ohe Bay, Hawai'i" PI: Travis Courtney	\$1,985
2015	National Science Foundation Graduate Research Fellowship Program PI: Travis Courtney	\$138,000

Peer-reviewed publications: Mentored Student in Biogeochemistry and Ecology Research Group

48. Toth LT and **Courtney TA**. 2026. Repeated coral bleaching events have eroded reef accretion potential in the Florida Keys. *Frontiers in Marine Science* 13:1878462. <https://doi.org/10.3389/fmars.2026.1878462>
47. Kekuewa SAH, Rintoul MS, **Courtney TA**, Pezner AK, Takeda T, Kayanne H, Mitarai S, Andersson AJ. 2026. Seasonal Reef-Scale Variability in Seawater CO₂ Chemistry in the Presence and Absence of Seaweed

Cultivation in Onna, Okinawa, Japan. *Estuaries and Coasts* 49(168). <https://doi.org/10.1007/s12237-026-01770-x>

46. Cordero E, Sherman CE, Yadav P, Kumar Raju V, Townsend JE, **Courtney TA**. 2026. Coral Intel: a YOLOv8 deep learning framework for monitoring Caribbean corals. *Intelligent Marine Technology and Systems* 4(18). <https://doi.org/10.1007/s44295-026-00107-5>
45. Infante R, Centeno L, **Courtney TA**, Cruz Motta JJ, Armstrong RA. 2026. Quantifying Marine Surface Microplastics in La Parguera Natural Reserve, Puerto Rico. *Microplastics* 5(108). <https://doi.org/10.3390/microplastics5020108>
44. DeCarlo TM, Jasnos O, Strange A, Andersson A, Bautista VII AT, Bloomer S, Bolden IW, Bosman M, Brachert TC, Braz GB, Cardoso GO, Carricart-Ganivet JP, Carilli JE, Castillo KD, Cavole L, Chan S, Chen X, Chomitz B, Correge T, **Courtney TA**, Deigan M, D'Olivo JP, Dunbar R, Enochs IC, Falsarella L, Felis T, Gutierrez-Estrada G, Hedger B, Hu S, Jameson S, Jupiter S, Kench P, Kersting DK, Lin K, Liu Y-W, Lorigados CAB, Manzello DP, McCulloch MT, Mies M, Moura RL, Oberle F, Pereira N, Prouty N, Ramos RD, Ren H, Ryan E, Thompson DM, Toth LT, Vergotti MJ, Webster JM, Zinke J. 2026. CoralCache: a virtual coral core repository for transparent and reproducible annual growth rate analyses, *Earth System Science Data* 18, 3341–3353. <https://doi.org/10.5194/essd-18-3341-2026>
43. Kekuewa SAH, Pezner AK, **Courtney TA**, Wei Y, Rintoul MS, Chou W-C, Soong K, Andersson AJ. 2026. Present and future seawater CO₂ chemistry across multiple coral reef habitats and scales at Dongsha Atoll and Taiping Island in the South China Sea. *Coral Reefs*. <https://doi.org/10.1007/s00338-026-02868-1>
42. Toth LT, Borer ET, Burkepile DE, Dudley J, Lemoine NP, Renzi JJ, Smith KEL, **Courtney TA**, Goeking SA, Hammond WM, Hoover DL, MacFayden S, Osland MJ, Townsend JE, Fidler RY. 2026. Scenarios and strategies for future-proofing ecosystem management under climatic novelty. *Conservation Biology*. e70250. <https://doi.org/10.1111/cobi.70250>
41. Cornwall CE, Timmerman O, Andersson A, Berbece DM, Branson O, Comeau S, **Courtney TA**, Donner SD, Enochs I, Harvey BP, Isaak AL, Koch HE, Logan CA, Perry CT, Schoepf V, Toth LT, Webb A. 2026. Persistence of coral reef structures into the twenty-first century. *Nature Reviews Earth & Environment*. 2026 Feb 19:1-1. <https://doi.org/10.1038/s43017-026-00764-4>
40. **Courtney TA**, Ricaurte-Chica M, Bayo Torres AD, Chabrier-Alpi AM, Medina Martinez AG, Melendez Declet CV, Mejias-Rivera CL, Fontanez Diaz CF, Hernández Rodríguez C, Lebron-Maldonado CI, Diaz Jimenez E, Irizarry Torres I, Carballeira Martínez J, Brito-Vera MJ, Luquis Ramos I, Santos Echevarría I, Collazo Reyes JJ, Centeno Mejias LJ, Nieves-Ortiz MA, Olmeda Saldaña MF, Gordon M, Mendez Falcon V, Reyes Claudio VR, Garcia Fernandez A, Melendez Aguilar AR, Garcia Hernandez WE, Feliberty Ruiz A, Cruz-Motta JJ. Puerto Rico Coral Reef Monitoring Program Water Quality Data from 2023–2025. 2026. *Scientific Data* 13(150). <https://doi.org/10.1038/s41597-025-06468-6>
39. Luquis Ramos IM, Williams SM, Cruz-Motta JJ, **Courtney TA**. 2026. Modeling bioerosion rates by *Diadema antillarum* populations to improve coral reef restoration in the Caribbean. *Restoration Ecology*. <https://doi.org/10.1111/rec.70302>
38. Rodríguez Hernández C, Reyes JC, Perez DI, Gonzalez-Corredor J, Krause JR, Campbell JE, Fourqurean JW, Cruz Motta JJ, **Courtney TA**. 2026. Influence of Environmental Factors on pH Buffering, Productivity, and Blue Carbon Storage in Contrasting *Thalassia testudinum* Seagrass Meadows in La Parguera, Puerto Rico. *Estuaries and Coasts* 49(40). <https://doi.org/10.1007/s12237-025-01652-8>
37. Mejias-Rivera CL, Ricaurte-Chica M, Carballeira-Martínez J, Hernández WJ, Armstrong RA, Cruz-Motta JJ, **Courtney TA**. 2026. Assessment of Coral Reef Eutrophication Thresholds Across Puerto Rico Using In-situ and Satellite-derived Chlorophyll-*a*. *Estuaries and Coasts* 49(26). <https://doi.org/10.1007/s12237-025-01644-8>
36. Martínez Ortiz J, Perez Perez J, Armstrong RA, Cruz Motta JJ, **Courtney TA**. 2025. Hypoxia driven by Caribbean Sargassum accumulation events. *Royal Society Open Science* 12(8):250322. <https://doi.org/10.1098/rsos.250322>
35. **Courtney TA** & Andersson AJ. 2025. Revisiting the terms used in net ecosystem calcification studies, carbonate budgets, and the equation of coral reef growth. *Coral Reefs* 44: 1031–1036. <https://doi.org/10.1007/s00338-025-02643-8>

34. Sanassy Pilly S, Townsend JE, Alisa CA, Razak TB, Roche RC, Turner JR, Chan S, Kriegman DJ, Andersson AJ, Perry CT, Lange ID, **Courtney TA**. 2025. Quantifying coral reef carbonate budgets: a comparison between *ReefBudget* and *CoralNet*. *Coral Reefs* 44: 513–527. <https://doi.org/10.1007/s00338-025-02620-1>
33. Pezner AK, **Courtney TA**, Chou W-C, Chu H-C, Frable BW, Kekuewa SAH, Soong K, Wei Y, Andersson AJ. 2024. Coral growth along a natural gradient of seawater temperature, pH, and oxygen in a nearshore seagrass bed on Dongsha Atoll, Taiwan. *PLoS ONE* 19(10): e0312263. <https://doi.org/10.1371/journal.pone.0312263>
32. Mejias-Rivera CL & **Courtney TA**. 2024. Ocean Warming, Heat Stress, and Coral Bleaching in Puerto Rico. *Caribbean Journal of Science* (54): 132-149. <https://doi.org/10.18475/cjos.v54i1.a17>
31. Toth LT, **Courtney TA**, Colella M, Ruzicka RR. 2023. Stony coral tissue loss disease accelerated shifts in coral composition and declines in reef accretion potential in the Florida Keys. *Frontiers in Marine Science*. <https://doi.org/10.3389/fmars.2023.1276400>
30. Mejias-Rivera CL, Armstrong RA, Balint S, García-Troche E, McKinney RA, Morell JM, Oczkowski A, **Courtney TA**. 2023. Localized inshore warming, acidification, and elevated particulate organic matter across a coupled mangrove, seagrass, and coral reef ecosystem in La Parguera, Puerto Rico. *Coral Reefs*. <https://doi.org/10.1007/s00338-023-02435-y>
29. Cornwall CE, Carlot J, Branson O, **Courtney TA**, Harvey BP, Perry CT, Andersson AJ, Diaz-Pulido G, Johnson MD, Kennedy E, Krieger EC, Mallela J, McCoy SJ, Nugues MM, Quinter E, Ross CL, Ryan E, Saderne V, Comeau S. 2023. Crustose coralline algae can contribute more than corals to coral reef carbonate production. *Nature Communications Earth & Environment*. <https://doi.org/10.1038/s43247-023-00766-w>
28. Pezner AK, **Courtney TA**, Barkley HC, Chou WC, Chu HC, Clements SM, Cyronak T, DeGrandpre MD, Kekuewa SAH, Kline DI, Liang Y-B, Martz TB, Mitarai S, Page HN, Rintoul MS, Smith JE, Soong K, Takeshita Y, Tresguerres M, Wei Y, Yates KK, Andersson AJ. 2023. Increasing hypoxia on global coral reefs under ocean warming. *Nature Climate Change* 13, 403–409 <https://doi.org/10.1038/s41558-023-01619-2>
27. Rintoul M, **Courtney T**, Dohner J, Giddings S, Inoha K, Kekuewa S, Mitarai S, Monismith S, Pezner A, Andersson A. 2022. The Effects of Light Intensity and Flow Speed on Biogeochemical Variability within a Fringing Coral Reef in Onna-son, Okinawa, Japan. *Journal of Geophysical Research – Oceans*. <https://doi.org/10.1029/2021JC018369>
26. Kekuewa SAH, **Courtney TA**, Cyronak T, Andersson AJ. 2022. Seasonal nearshore ocean acidification and deoxygenation in the Southern California Bight. *Scientific Reports*. <https://doi.org/10.1038/s41598-022-21831-y>
25. Toth LT, **Courtney TA**, Colella MA, Kupfner Johnson SA, Ruzicka RR. 2022. The past, present, and future of coral reef growth in the Florida Keys. *Global Change Biology*. <https://doi.org/10.1111/gcb.16295>
24. **Courtney TA**, Barkley HC, Chan S, Couch CS, Kindinger TL, Oliver TA, Kriegman DJ, Andersson AJ. 2022. Rapid assessments of Pacific Ocean net coral reef carbonate budgets and net calcification following the 2014-2017 global coral bleaching event. *Limnology and Oceanography*. <https://doi.org/10.1002/lno.12159>
23. Elahi R, Edmunds PJ, Gates RD, Kuffner IB, Barnes BB, Chollett I, **Courtney TA**, Guest JR, Lenz EA, Toth LT, Viehman TS, Williams ID. 2022. Scale dependence of coral reef oases and their environmental correlates. *Ecological Applications*. <https://doi.org/10.1002/eap.2651>
22. Bresnahan P, Cyronak T, Brewin RJW, Andersson A, Wirth T, Martz T, **Courtney T**, Hui N, Kastner R, Stern A, McGrain T, Reinicke D, Richard J, Hammond K, Waters S. 2022. A High-Tech, Low-Cost, Internet of Things Surfboard Fin for Coastal Citizen Science, Outreach, and Education. *Continental Shelf Research*. <https://doi.org/10.1016/j.csr.2022.104748>
21. **Courtney TA**, Cyronak T, Griffin AJ, Andersson AJ. 2021. Implications of salinity normalization of seawater total alkalinity in coral reef metabolism studies. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0261210>
20. Kekuewa SAH, **Courtney TA**, Cyronak T, Kindeberg T, Eyre BD, Stoltenberg L, Andersson AJ. 2021. Temporal and Spatial Variabilities of Chemical and Physical Parameters on the Heron Island Coral Reef Platform. *Aquatic Geochemistry*. <https://doi.org/10.1007/s10498-021-09400-7>

19. **Courtney TA**, Guest JR, Edwards AJ, Dizon RM. 2021. Linear extension, skeletal density, and calcification rates of the blue coral *Heliopora coerulea*. *Coral Reefs*. <https://doi.org/10.1007/s00338-021-02137-3>
18. Pezner AK, **Courtney TA**, Page HN, Giddings SN, Beatty CM, DeGrandpre MD, Andersson AJ. 2021. Lateral, Vertical, and Temporal Variability of Seawater Carbonate Chemistry at Hog Reef, Bermuda. *Frontiers in Marine Science*. <https://doi.org/10.3389/fmars.2021.562267>
17. **Courtney TA**, Kindeberg T, Andersson AJ. 2020. Coral calcification responses to the North Atlantic Oscillation and coral bleaching in Bermuda. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0241854>
16. Kindeberg T, Bates NR, **Courtney TA**, Cyronak T, Griffin A, Mackenzie FT, Paulsen M-L, Andersson AJ. 2020. Porewater Carbonate Chemistry Dynamics in a Temperate and a Subtropical Seagrass System. *Aquatic Geochemistry*. <https://doi.org/10.1007/s10498-020-09378-8>
15. **Courtney TA**, Barnes BB, Chollett I, Elahi R, Gross K, Guest JR, Kuffner IB, Lenz EA, Nelson HR, Rogers CS, Toth LT, Andersson AJ. 2020. Disturbances drive changes in coral community assemblages and coral calcification capacity. *Ecosphere*. <https://doi.org/10.1002/ecs2.3066>
14. Cyronak T, Takeshita Y, **Courtney TA**, DeCarlo EH, Eyre BD, Kline DI, Martz T, Page H, Price NN, Smith J, Stoltenberg L, Tresguerres M, Andersson AJ. 2020. Diel temperature and pH variability scale with depth across diverse coral reef habitats. *Limnology and Oceanography Letters*. <https://doi.org/10.1002/lol2.10129>
13. Baumann JH, Ries JB, Rippe JP, **Courtney TA**, Aichelman HE, Westfield I, Castillo KD. 2019. Nearshore coral growth declining on the Mesoamerican Barrier Reef System. *Global Change Biology*, 25: 3932–3945. <https://doi.org/10.1111/gcb.14784>
12. **Courtney TA** & Andersson AJ. 2019. Evaluating measurements of coral reef net ecosystem calcification rates. *Coral Reefs*, 38(5):997–1006. <https://doi.org/10.1007/s00338-019-01828-2>
11. Page HN, **Courtney TA**, De Carlo EH, Howins NM, Koester I, Andersson AJ. 2019. Spatiotemporal variability in seawater carbon chemistry for a coral reef flat in Kāne'ohe Bay, Hawai'i. *Limnology and Oceanography*, 63:913–934. <https://doi.org/10.1002/lno.11084>
10. Guest JR, Edmunds PJ, Gates RD, Kuffner IB, Andersson AJ, Barnes BB, Chollett I, **Courtney TA**, Elahi R, Gross K, Lenz EA, Mitarai S, Mumby PJ, Nelson HR, Parker BA, Putnam HM, Rogers CS, Toth LT. 2018. A framework for identifying and characterising coral reef “oases” against a backdrop of degradation. *Journal of Applied Ecology*, 00:1-11. <https://doi.org/10.1111/1365-2664.13179>
9. **Courtney TA**, De Carlo EH, Page HN, Bahr KD, Barro A, Howins N, Tabata R, Terlouw G, Rodgers KS, Andersson AJ. 2018. Recovery of reef-scale calcification following a bleaching event in Kāne'ohe Bay, Hawai'i. *Limnology & Oceanography Letters*, 3:1–9. <https://doi.org/10.1002/lol2.10056>
8. **Courtney TA**, Lebrato M, Bates NR, Collins A, de Putron SJ, Garley R, Johnson, R, Molinero JC, Noyes TJ, Sabine CL, Andersson AJ. 2017. Environmental controls on modern scleractinian coral and reef-scale calcification. *Science Advances*, 3(11), p.e1701356. <https://doi.org/10.1126/sciadv.1701356>
7. Page HN, **Courtney TA**, Collins A, De Carlo EH, Andersson AJ. 2017. Net community metabolism and seawater carbonate chemistry scale non-intuitively with coral cover. *Frontiers in Marine Science*, 4:161. <https://doi.org/10.3389/fmars.2017.00161>
6. **Courtney TA**, Andersson AJ, Bates NB, Collins A, Cyronak T, de Putron SJ, Eyre BD, Garley R, Hochberg EJ, Johnson R, Musielewicz S, Noyes T, Sabine CL, Sutton AJ, Tancin J, Tribollet A. 2016. Comparing Chemistry and Census-based Estimates of Net Ecosystem Calcification on a Rim Reef in Bermuda. *Frontiers in Marine Science*, 3:181. <https://doi.org/10.3389/fmars.2016.00181>
5. Baumann JH, Townsend JE, **Courtney TA**, Aichelman HE, Davies SW, Lima FP, Castillo KD. 2016. Temperature Regimes Impact Coral Assemblages along Environmental Gradients on Lagoonal Reefs in Belize. *PLoS ONE*, 11(9): e0162098. <https://doi.org/10.1371/journal.pone.0162098>
4. Aichelman HE, Townsend JE, **Courtney TA**, Baumann JH, Davies SW, Castillo KD. 2016. Heterotrophy mitigates the response of the temperate coral *Oculina arbuscula* to temperature stress. *Ecology and Evolution*, 6(18): 6758-6769. <https://doi.org/10.1002/ece3.2399>
3. Horvath KM, Castillo KD, Armstrong P, Westfield IT, **Courtney T**, Ries JB. 2016. Next-century ocean acidification and warming both reduce calcification rate, but only acidification alters skeletal morphology of reef-building coral *Siderastrea siderea*. *Scientific Reports*, 6:29613. <https://doi.org/10.1038/srep29613>

2. **Courtney T** and Ries JB. 2015. Impact of atmospheric pCO₂, seawater temperature, and calcification rate on the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ composition of echinoid calcite (*Echinometra viridis*). Chemical Geology, 411: 228-239. <https://doi.org/10.1016/j.chemgeo.2015.06.030>
1. **Courtney T**, Ries JB, Westfield I. 2013. Predicted end of 21st century CO₂-induced ocean acidification decreases calcification rates in the tropical urchin *Echinometra viridis*. Journal of Experimental Marine Biology and Ecology, 440: 169-175. <https://doi.org/10.1016/j.jembe.2012.11.013>

Computational tools and datasets:

3. **Courtney TA**, Lange ID, Sannassy Pilly, S, Townsend JE, Chan S, Perry CT, Kriegman DJ, Andersson AJ. 2024. Area-normalized scaling of ReefBudget calcification, macrobioerosion, and microbioerosion rates for use with CoralNet Version 2.0. Zenodo. <https://doi.org/10.5281/zenodo.13257132>
3. **Courtney TA**, Andersson AJ. 2021. Calcification Dissolution Potential Tool for Excel: Version 1. Zenodo. <https://doi.org/10.5281/zenodo.7051628>
2. Chan S, **Courtney TA**, Andersson AJ, Kriegman DJ. 30 July 2021. "CoralNet now estimates carbonate production rates." CoralNet. <https://coralnet.ucsd.edu/blog/coralnet-now-estimates-carbonate-production-rates/>
1. **Courtney TA**, Chan S, Lange ID, Perry CT, Kriegman DJ, Andersson AJ. 2021. Area-normalized scaling of ReefBudget calcification, macrobioerosion, and microbioerosion rates for use with CoralNet Version 1.0. Zenodo. <https://doi.org/10.5281/zenodo.5140477>

Conference presentations and invited talks:

28. **Courtney TA**. 2026. Developing and Validating Automated Tools to Assess Coral Reef Carbonate Production. University of Puerto Rico Mayagüez Department of Geology Annual Symposium.
27. **Courtney TA**. 2025. Coral bleaching and coastal hypoxia in Puerto Rico. University of Puerto Rico Mayagüez Department of Biology Seminar invited speaker.
26. **Courtney TA** & Cruz Motta JJ 2025. Monitoring water quality, fish and corals in Puerto Rico's coral reefs and applications of the Biological Condition Gradient (BCG). Puerto Rico Department of Natural and Environmental Resources.
25. **Courtney TA**. 2023. Coral reef calcification and biogeochemistry under environmental change. University of Puerto Rico Mayagüez Department of Chemistry Seminar invited speaker.
24. **Courtney TA**. 2022. Blue carbon uptake and buffering against ocean acidification by seagrasses. EcoEléctrica invited talk.
19. **Courtney TA**. 2022. Benthic metabolism and buffering against ocean acidification. University of Puerto Rico Mayagüez – Purdue University Blue Initiative Partnership Meeting.
23. **Guest lecture**. 2021. Ocean Briefs Feedback and Discussion for University of Montana Story Lab
22. **Courtney TA** & Andersson AJ. 2021. Tools to assess coral reef calcification. NOAA Ocean Acidification Working Group Meeting.
21. **Courtney TA**. 2021. Coral reef calcification under rapid environmental change. AECiMA Outreach Talks: University of Puerto Rico, Mayagüez.
20. **Courtney TA**. 2021. Coral reef metabolism and biogeochemistry under rapid environmental change. University of Puerto Rico, Mayagüez.
19. **Guest lecture**. 2020. "Disturbances and coral reef calcification" for NSF Summer REU Program
18. **Guest lecture**. 2020. Rapid estimates of coral reef calcification for SIO 119 Undergraduate Course
17. **Guest lecture**. 2020. "The chemistry (and physics) of quantifying net coral reef calcification" for SIO 119
16. **Courtney TA**. 2020. Disturbances drive changes in coral community assemblages and coral calcification capacity (and the development of tools to assess reef-scale calcification under OA). NOAA Ocean Acidification Community Meeting.
15. **Courtney TA**. 2019. Coral calcification and climate change. Scripps Education Association.
14. **Courtney TA** and Andersson AJ. 2019. Evaluating measurements of ecosystem-scale coral reef calcification under global environmental change. ASLO Aquatic Sciences Meeting.

13. **Guest lecture.** 2019. "Marine Chemistry Basics" for SIO Master of Advanced Studies
12. **Guest lecture.** 2019. Paleoclimatology Lab for SIO Master of Advanced Studies (with Dr. Art Miller)
11. **Courtney TA.** 2018. The science (and value) of coral reef calcification. San Diego State Environmental Business Society.
10. **Courtney TA.** 2018. Environmental controls on coral and reef-scale calcification. National Sun Yat-sen University lunch seminar.
9. **Courtney TA, Lebrato M, Bates NR, Collins A, de Putron SJ, Garley R, Johnson, R, Molinero JC, Noyes TJ, Sabine CL, Andersson AJ.** 2018. New insights into the drivers of coral and reef-scale calcification from Bermuda. Ocean Sciences Meeting.
8. **Courtney TA.** 2017. Environmental controls on coral and reef-scale calcification. San Diego Coral Club.
7. **Courtney TA, Andersson AJ, De Carlo EH, Page HN, Koester I, Terlouw G, Tabata R, Bahr KD, Rodgers KS.** 2017. Coral bleaching impacts on reef-scale net calcification and net community production in Kāne'ohe Bay, HI. ASLO Ocean Sciences Meeting.
6. **Courtney TA and Andersson AJ.** 2016. Seasonal patterns in calcium carbonate production of a Bermuda coral reef. Scripps Student Symposium.
5. **Courtney TA, Andersson AJ, Cyronak T, Noyes T, Bates NR, Collins A, de Putron S, Garley R, Hochberg EJ, Johnson R.** 2016. Comparing chemistry and census-based estimates of net ecosystem calcification on a rim reef in Bermuda. 13th International Coral Reef Symposium.
4. **Courtney T, Baumann J, Foguel AD, Horvath K, Westfield I, Castillo KD, Ries JB.** 2014. Characterizing 21st century growth trends of the scleractinian coral *Siderastrea siderea* throughout the Belize barrier reef and atoll system. Benthic Ecology Meeting.
3. **Courtney T, Ries JB, Westfield I.** 2013. Impact of atmospheric pCO₂ and seawater temperature on calcification rate and stable isotopic composition ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$) of echinoid calcite (*Echinometra viridis*). UNC at Chapel Hill Anadarko Research Symposium.
2. **Courtney T, Ries JB, Westfield I.** 2012. *Echinometra viridis* exhibits seasonal response in calcification rates to predicted end of 21st century CO₂-induced ocean acidification. Geological Society of America Fall Meeting.
1. **Courtney T, Ries JB, Westfield I.** 2012. Effects of warming and CO₂-induced acidification on the tropical urchin *Echinometra viridis*. UNC at Chapel Hill Department of Marine Sciences Seminar Series.

Fieldwork experience:

2025	La Parguera, Puerto Rico: Monthly seawater chemistry sampling
2024	La Parguera, Puerto Rico: Coral reef benthic surveys, instrument deployments
2023	Hawai'i: E/V Nautilus Cruise NA 156 - Ocean Exploration Through Advanced Imagery
2023	La Parguera, Puerto Rico: Seagrass chemistry surveys, instrument deployments, cores, and productivity
2019	Okinawa, Japan: Chemistry surveys, instrument deployments, and lab experiments
2019	Taiping Island, Taiwan (lead): Chemistry surveys, instrument deployments, and benthic surveys
2018	Bermuda (lead): MAPCO2 buoy servicing
2018	Dongsha Atoll, Taiwan: Chemistry surveys, porewaters, instrument deployments, and coral coring
2017	Bermuda: MAPCO2 buoy servicing, seawater chemistry surveys, and instrument deployments
2017–2021	La Jolla Bight, California: Chemistry surveys and instrument deployments
2017–2018	Mission Bay, California: Chemistry surveys, porewaters, and instrument deployments
2017	Kāne'ohe Bay, Hawai'i (lead): Chemistry surveys
2016	Kāne'ohe Bay, Hawai'i: Chemistry surveys, porewaters, and instrument deployments
2016	Bermuda: MAPCO2 buoy servicing and coral coring
2016	Kāne'ohe Bay, Hawai'i: Chemistry surveys, benthic surveys, and mesocosm experiments
2014	Belize Barrier Reef System: Coral reef community benthic surveys
2014	Southern Outer Banks, North Carolina: Coral sample collection for lab experiments
2013	Southern Outer Banks, North Carolina: Benthic habitat surveys
2012	Belize Barrier Reef System: Coral coring

Teaching experience:

2026 Spring	Instructor: Chemical Oceanography, University of Puerto Rico Mayagüez
2025 Fall	Instructor: Open Data Science in Chemical Oceanography, University of Puerto Rico Mayagüez
2025 Spring	Instructor: Chemical Oceanography, University of Puerto Rico Mayagüez
2024 Fall	Instructor: Marine Pollution, University of Puerto Rico Mayagüez
2024 Spring	Instructor: Chemical Oceanography, University of Puerto Rico Mayagüez
2023 Fall	Instructor: Marine Pollution, University of Puerto Rico Mayagüez
2023 Spring	Instructor: Chemical Oceanography, University of Puerto Rico Mayagüez
2023 Spring	Instructor: Chemical Oceanography Data Analysis Laboratory, University of Puerto Rico Mayagüez
2022 Fall	Instructor: Marine Pollution, University of Puerto Rico Mayagüez
2022 Spring	Instructor: Chemical Oceanography, University of Puerto Rico Mayagüez
2022 Spring	Instructor: Chemical Oceanography Data Analysis Laboratory, University of Puerto Rico Mayagüez
2021 Fall	Instructor: Marine Pollution, University of Puerto Rico Mayagüez
2021 Fall	Instructor: Professional Ethics in Marine Sciences, University of Puerto Rico Mayagüez
2018	Instructor: SIO 90 Perspectives on Ocean Science
2013	Teaching Assistant: GEOL 101L Introductory Geology Lab, UNC

Graduate students mentored:

University of Puerto Rico at Mayagüez Committee Chair:

2025–present	Kenneth Montalvo Negron, MS, Scaling up seaweed production in the Caribbean
2025–present	Amanda Melendez Perez, MS, Metabolism and carbonate production of soft corals
2025–present	Alejandra Aguilar-Giraldo, PhD, Quantifying rates and drivers of coral reef bioerosion
2023–present	Joseph Townsend, PhD, Rates, drivers, and restoration of coral reef carbonate production
2024–2026	Janluis Rivera, MS, Modeling wave attenuation by coral restoration (co-chair: Dr. Andre Amador)
2023–2026	Carolina Melendez Declet, MS, Designing integrated multitrophic aquaculture for <i>Ulva</i> seaweed
2023–2025	Ignacio Rueda, MS, Photogrammetry to quantify structural complexity provisioning by deep-sea corals
2022–2025	Juanita Carballeira Martinez, MS, Vertical, spatial, and temporal variability in coral reef biogeochemistry
2022–2025	Carla Mejías Rivera, PhD, Remote sensing and in situ assessments of water quality in Puerto Rico
2022–2025	Irais Luquis Ramos, MS, Modeling <i>Diadema</i> sea urchin bioerosion to inform restoration strategies
2022–2024	Jose Martinez Ortiz, MS, Sargassum accumulation events as drivers of Caribbean coastal hypoxia
2022–2024	Catherine Hernández Rodriguez, MS, Seagrass carbon cycling, pH buffering, and carbon sequestration
2021–2023	Leira Centeno Mejias, MS, Water quality monitoring compliance project

Scripps Institution of Oceanography Informal Mentoring:

2019–2021	Ariel Pezner, PhD, Impacts of hypoxia on coral reefs
2019–2021	Sam Kekuewa, PhD, Spatiotemporal variability of seawater chemistry in coastal ecosystems
2018–2019	Thompson Banez, MAS in Marine Biodiversity and Conservation Student Mentor
2017–2018	Emily Parker, MAS in Marine Biodiversity and Conservation Student Mentor
2017–2018	Sam Kekuewa, MS Graduate Peer Mentor Program
2017–2018	Wiley Wolfe, PhD Graduate Peer Mentor Program

Undergraduate students mentored:

2026-present	Sediment characterization relative to coral reef carbonate production (UPRM): Ariana Negron Pacheco
2025-2026	Segmentation of large-area coral reef imagery (UPRM): Paola Berrios Morales, Ariana Negron Pacheco
2023-2024	Analysis of seawater quality (UPRM): Victoria Reyes Claudio, Claudia Lebron Moldonado
2023	Blue Integrated Partnerships Summer Internship Program (UPRM): Alexis Warnecke & Victoria Reyes Claudio *Awarded 1 st Place for Best Presentation
2020–2021	Seawater carbonate chemistry and inshore-offshore gradients of coral reef biogeochemistry (UCSD): William Tallentire
Summer 2020	Analysis of coral cover following coral bleaching, NSF Research Experiences for Undergraduates (UCSD): Audrey Ellias Zach Ferris

2013–2014	Coral skeletal geochemistry, growth rate analysis, and marine aquarium facilities (UNC-CH):		
	Hannah Aichelman	Courtney Anderson	Madelyn Roycroft
	Pualani Armstrong	Jessica Boulton	Joseph Townsend
	Carissa Campbell	Kathryn Cobleigh	Vallari Eastman
	Ashley Foguel	Hannah Knight	

Graduate student committees:

2026–present	Angel Acosta Colon, UPRM PhD in Marine Sciences
2026–present	Kevin Martinez Lopez, UPRM MS in Marine Sciences
2026–present	Mariela Cortes Medina, UPRM MS in Marine Sciences
2026–present	Carolina Cesar Avila, UPRM MS in Marine Sciences
2025–present	Jomar Camacho Garay, UPRM MS in Marine Sciences
2025–present	Melanie Cordero Cruz, UPRM MS in Marine Sciences
2023–present	Jenniffer Perez Perez, UPRM PhD in Marine Sciences
2023–present	Adiana Bayo Torres, UPRM MS in Marine Sciences
2023–present	Maria Jose Brito Vera, UPRM PhD in Marine Sciences
2025–2026	Glorimar Franqui Rivera, UPRM PhD in Marine Sciences
2024–2026	Andrea Velez Rivera, UPRM MS in Marine Sciences
2024–2026	Biajani Gonzalez Torres, UPRM MS in Marine Sciences
2022–2025	Raymond Infante Rosa, UPRM MS in Marine Sciences
2022–2024	Ana Medina Martinez, UPRM MS in Marine Sciences

Working groups, workshops, and field courses:

2024 & 2025	Drivers of ecological stability in response to climate change across ecosystems, USGS Powell Center
2024	Puerto Rico Coral Reef Restoration Action Plan, ISER Caribe, NOAA, DRNA, and The Nature Conservancy
2023	Caribbean Carbon Accounting in Seagrass, Florida International University and The Nature Conservancy
2023	D-ENTERPRISE Conference to Increase Diversity in the Ocean Sciences, Office of Naval Research
2022	Will coral reefs become built by calcifying seaweed in the future?, French Embassy in New Zealand
2017	Local-scale coral reef resilience under global-scale ocean change, USGS Powell Center
2017	Coral In Situ MEtabolism (CISME) workshop, Kāne'ohe Bay, Hawai'i Institute of Marine Biology
2016	Natural History Below the Tides, La Jolla, Scripps Institution of Oceanography
2012	Field Geology in the Rocky Mountains, Montana, Judson Mead Geologic Field Station
2011	Human and Marine Ecology, Galápagos Islands, Universidad San Francisco de Quito
2011	Marine Resources Population Dynamics Workshop, NOAA NMFS
2010	Alternative Fall Break Environmental Trip Co-Leader: UNC APPLES Service Learning

Outreach and service:

2024	Panelist on NSF GRFP for UNC Chapel Hill Coral Reefs and Climate Change course
2023	Ocean Exploration with Dr. Bob Ballard Livestream for University of Puerto Rico
2020	SciREN scientist participant, created lesson plan on coral reef growth for K-12 teachers
2020–2021	Mentor for Científico Latino Graduate Student Mentorship Initiative
2019	University of California delegate to the 25 th United Nations Conference of Parties
2019–2021	Letters to a Pre-Scientist participant
2018	Panelist for San Diego State University Earth Week Chasing Coral screening
2018	Ocean Acidification lecture for UCSD Retirement Association
2018	Panelist for UCSD Retirement Association Chasing Coral screening
2017	Panelist for Citizens Climate Lobby Chasing Coral screening
2017	Panelist for Smartfin + Changing Tides Chasing Coral screening
2015–2021	Smartfin project surfboard fin sensors β–tester
2015–2020	Scripps Community Outreach Program for Education (SCOPE) volunteer

2015–2016 Rosa Parks Elementary School volunteer tutor
 2015, 16, 18 Ocean Discovery Institute volunteer scientist
 2014 SciREN scientist participant, created lesson plan on ocean acidification for K-12 teachers
 2012 NC Museum Natural Sciences Marine Mammal Day volunteer

Job related certifications:

2021 Virtual Teacher Certificate, University of California Irvine
 2020–present Freediving Instructors International Level 1 Freediver
 2020 Introduction to College Teaching, University of California San Diego
 2018–present California Boater License
 2017–present Motorboat Operator Certification Course
 2011–present AAUS Science Diver to 60 ft. + nitrox (>200 dives)
 2011–present First Aid, CPR, Emergency O₂, and Diving Neurological Assessment Certified
 2010–present North Carolina Boater Education Certified

Diversity, Equity, and Inclusion Trainings

2023 Harassment and Discrimination Prevention for Non-Supervisors, EasyLlama
 2020 Introduction to College Teaching, UC San Diego Teaching + Learning Commons
 2020 Self-Guided Foundational Safe Zone Training, Safe Zone Project
 2020 Transfer Ally Training, UC San Diego
 2020 Undocu-Ally Training, UC San Diego
 2020 Conflict de-escalation Training, Hollaback!

Academic service:

2025–Present UPRM Department of Marine Sciences – Curriculum Committee Member
 2024–Present UPRM College of Arts and Sciences – Dialogue Committee Member
 2022–Present UPRM Department of Marine Sciences – Graduate Committee Chair
 2021–Present UPRM Department of Marine Sciences – Appraisal Committee Member
 2021–2024 UPRM Department of Marine Sciences – Diving Control Board Member
 2022 National Science Foundation Review Panelist
 2021 Sea Grant Puerto Rico Technical Review Panelist
 2019 Session co-chair: Coral Reef Ecosystems, ASLO Aquatic Sciences Meeting
 2017–2019 Treasurer + Operations, Scripps Academic Club

Peer review:

Global Change Biology	Nature Climate Change
Hydrobiologia	Limnology and Oceanography
Proceedings of the Royal Society B	Coral Reefs
Marine Environmental Research	Bulletin of Marine Science
Geophysical Research Letters	PLOS ONE
National Science Foundation	Marine Chemistry
Marine Ecology Progress Series	One Earth
Hawai'i Sea Grant	Marine Ecology
Communications Earth & Environment	Scientific Reports
Palaeogeography, Palaeoclimatology, Palaeoecology	PeerJ
Journal of Experimental Marine Biology and Ecology	Frontiers in Marine Science
Regional Studies in Marine Science	Limnology and Oceanography: Methods
Science of the Total Environment	PNAS
Ecosphere	Global Biogeochemical Cycles
Marine & Freshwater Research	Limnology and Oceanography Letters
Earth Surface Processes and Landforms	